Signal Provenance in Retail Algorithmic Trading: An Architectural Gap in SEBI's 2026 Framework and Its General Form

Lokesh Gupta Founder, AION Analytics (India) July 2026

Abstract

SEBI's algorithmic trading framework, fully mandatory from 1 April 2026, closes the order chain: every algo order is tagged, every broker accountable, every provider empanelled. It cannot close the signal chain. This paper shows why, and that the gap is structural rather than a drafting failure.

A broker's order-placement interface receives an instruction to trade, symbol, quantity, price, side. It never receives what caused the instruction. No obligation imposed on a broker can capture what the broker is never sent. The same absence appears in MiFID II and RTS 6, in the VARA and DFSA rulebooks, and in the most rigorous order-chain audit standard identified in the field, each read against primary sources.

The consequence is visible in enforcement. Across four SEBI matters examined, signal chains were reconstructed after the fact, from screenshots, bank trails, and messaging records, at intervals of two to six years, and only after a complaint. Reconstruction is what a regulator does when observation is unavailable.

Production evidence gathered under the live mandate demonstrates that origin capture at the point a signal is received is technically feasible, on controlled infrastructure, without dependence on any single broker. The same mechanism permits a participant to set risk limits in rupees rather than quantitative parameters, and retains their financial data on their own device.

The paper closes by generalising: where a framework anchors accountability at an intermediary and the conduct of regulatory interest occurs upstream of it, the framework will reconstruct rather than observe. A diagnostic is proposed on that basis.

The same architecture that makes signal origin observable to the regulator makes risk controllable by the user and keeps their financial data private from everyone, closing the audit gap and the inclusion gap with one mechanism, and surrendering nothing to either broker or state.

Disclosure

The author is founder of AION Analytics (INDIA), which has developed a production system implementing the architecture described. A provisional patent covering the method has been filed. The findings in this paper were shared with relevant Indian authorities prior to publication.

1. Introduction

Algorithmic trading by retail participants is no longer an unregulated activity in most major markets. Over the past decade, regulators in India, the European Union, and elsewhere have built frameworks specifically to govern it: requiring firms to register, to maintain risk controls, to notify authorities of algorithmic activity, and to keep records sufficient to reconstruct who placed a given order and through what channel. Read across jurisdictions, the shape of this regulatory achievement is consistent. The chain from an order's arrival at a broker through to its execution, clearing, and settlement is, in every framework this paper examined, traceable to a specific, registered, accountable entity. Order-chain traceability, in this sense, is largely a closed question.

India's framework is a clear example of how thorough this closure can be. Beginning with a circular issued in February 2025 and sequenced through a deliberate, multi-milestone rollout, the framework reached full mandatory application for every stock broker in the country on 1 April 2026. It requires that every algorithmic order carry a unique, exchange-issued identifier; that brokers act as principal to the algo providers operating through their systems; that those providers be empanelled with the exchange before they may operate at all; and that access run over a whitelisted, static IP address specifically so that, in the regulator's own words, the algo provider and the end user can be identified and traced. This is not a thin or hastily assembled rulebook. It reflects sustained, careful regulatory work, and this paper treats it as such throughout.

And yet, examined against India's own enforcement record, a pattern recurs. Matters involving unregistered trading recommendations distributed through social media and messaging channels, cases this paper examines in detail, were resolved only after investigations lasting from roughly two and a half to more than six years, each built by reconstructing, after the fact, a chain of communications, bank transactions, and messaging-group records that traced back to when a recommendation was first made. In each case, the resulting order was itself fully compliant with the framework described above: properly tagged, placed through a registered channel, traceable to the account that placed it. What each investigation had to rebuild by hand was something the framework never asked any party to record in the first place, not who placed the order, but what caused it to be placed.

This paper asks why that pattern recurs, and argues that the answer is architectural rather than a shortcoming of any regulator's design. The interface a broker's order-placement system is built to receive is an instruction to trade, a symbol, a quantity, a price, a side. It is not built to receive, and no framework examined here requires it to receive, a record of the event that produced that instruction. This is not a gap any one regulator left open by oversight. It is a property of where the regulated interface sits in the chain from signal to execution, and it recurs, this paper finds, in every framework examined regardless of jurisdiction.

The scope of this paper is deliberately bounded. Its primary analysis is of India's regulatory framework, examined in full from primary sources, and of India's own enforcement record. Comparative observations from the European Union's MiFID II and RTS 6, and from the United Arab Emirates' VARA and DFSA rulebooks, are presented as findings from the specific provisions examined, not as a claim that the pattern holds universally across every jurisdiction or every asset class. Where those comparative findings rest on a narrower reading than the Indian analysis, this paper says so directly.

What follows is organised in three parts. Section 2 sets out the structural argument: what the regulatory record shows the order-chain closes, why the interface it regulates cannot extend to signal origin by construction, and what would be required to falsify that claim. Section 3 considers the consequences of the gap as it stands: the cost it imposes on enforcement, the strongest objections to treating it as significant, and a tension between India's stated ambition to broaden retail participation and one specific requirement of its own algorithmic trading framework. Section 4 presents production evidence that capturing signal origin, at the point a signal is received and before any order exists, is not merely a theoretical possibility but a demonstrated one, and examines whether the frictions identified in Section 3 are addressable at the same layer.

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2. The Structural Argument

Terminology, established against the strongest existing standard

Before the central claim is stated, one piece of prior art requires direct address, because it is the closest anyone has come to the problem this paper describes.

The VeritasChain Protocol (VCP) is, on present evidence, the most rigorous order-chain audit standard available. Its documentation states that it "records the complete lifecycle of every trading decision, from signal generation to execution completion," with the first stage of that lifecycle defined as "SIG (Signal), AI/Algorithm generates a trading decision." A reader encountering the phrase "signal generation" could reasonably conclude that the problem addressed here is already solved. It is not, and the reason reflects well on VCP's design rather than poorly. VCP's chain begins at the moment a trading system's own algorithm generates a decision. It does not extend to whatever external event caused that algorithm to generate the decision it did. VCP is mapped explicitly to MiFID II's RTS 6 and to the EU AI Act's logging requirements; it was built, correctly, to make exactly that boundary, a trading system's own decision point, auditable. That the most rigorous protocol built for this purpose stops precisely where this paper's concern begins is not evidence against the argument that follows. It is the strongest available evidence for it. If the discipline's own best instrument begins downstream of signal origin, the boundary is a property of the discipline, not an omission in any one implementation.

This paper uses "signal" to mean what precedes VCP's SIG stage: the external event, communication, or source that caused a trading decision to be generated, prior to and independent of the system that generated it. What follows examines one regulatory framework in full, against this definition: India's.

What SEBI's framework closes

SEBI's algorithmic trading framework was built deliberately and sequenced carefully. It deserves to be described in full, and on its own terms, before anything is said about its limits.

Circular SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/0000013, dated 4 February 2025, establishes the framework's core architecture. Brokers are made principal, with any algo provider or fintech vendor acting as their agent (§I(a)). Every algo order originating through a

broker's API must be tagged with a unique identifier issued by the exchange (§I(b)). Algo providers must be empanelled with the exchange before they may operate through a broker at all (§III(a)). Access is restricted to a unique, vendor-specific API key over a static, whitelisted IP address, stated in the circular's own words to "ensure identification and traceability of the algo provider and the end user." Providers of undisclosed, "black box" algorithms face a further requirement: registration as a Research Analyst, with a documented research report maintained for each algorithm and refreshed on any change to its logic (§V(a)(ii)).

A second circular, SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/132, dated 30 September 2025, sequenced the framework's rollout into a glide path rather than a single cutover, evidently in response to representations from brokers and vendors that immediate compliance was impractical. Registration applications were due by 31 October 2025, registration itself completed by 30 November 2025, and a mandatory mock trading session by 3 January 2026, with evidence of participation to be furnished to the exchange. Brokers who missed these milestones were barred, from 5 January 2026, from onboarding any new retail client onto API-based algo trading. The framework became fully mandatory, for every stock broker without exception, from 1 April 2026.

Taken together, this is a thorough and carefully staged regulatory achievement, built over more than a year of consultation, phased deliberately to allow the market to adapt rather than imposed abruptly. It closes the order chain. Every order that reaches an exchange under this framework can be traced to the algo provider that generated it, the broker that carried it, and the API key and IP address through which it travelled. On its own terms, and for the question it set out to answer, the framework works.

What the framework does not reach, and why

Every provision above answers one question: who placed this order, through which channel, under whose accountability. None of them answers what caused the order to be placed at all.

This reflects the shape of the instrument being regulated, not a limitation in how the framework was drafted. An order payload is an instruction to trade: symbol, quantity, price, side, product, validity. It is not a record of causation. A broker's function begins the moment that instruction arrives; whatever produced it, a recommendation received on a device, a signal generated by an algorithm, a message forwarded from elsewhere, occurred before the broker was party to the transaction in any capacity. A broker cannot log what it is never sent.

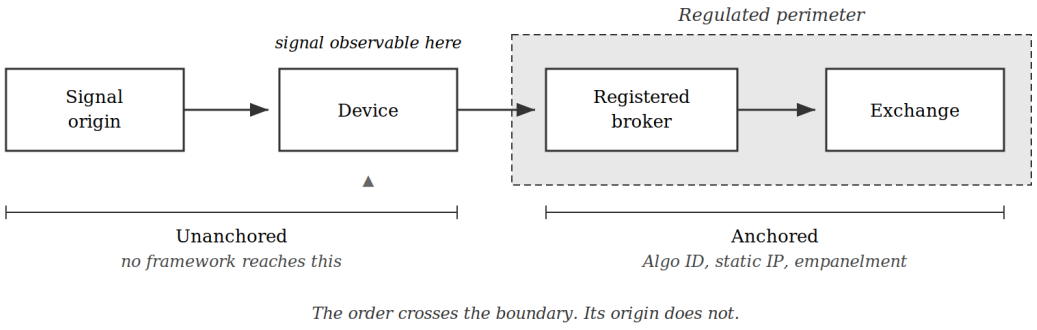


Figure 1. The order chain and the signal chain.

Figure 1. The order chain and the signal chain. Everything from the broker onward is identified, tagged, and traceable under the framework. Everything before it is not addressed. The order crosses the boundary; its origin does not.

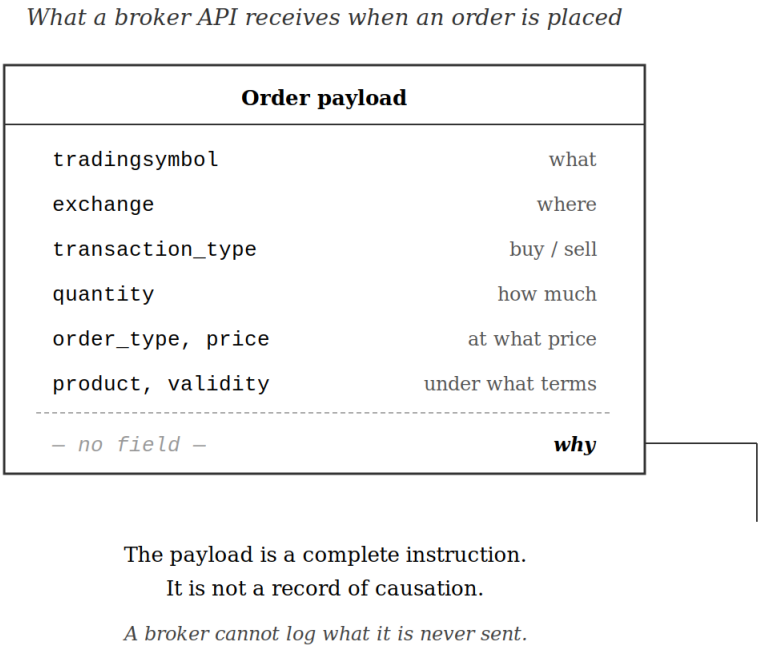


Figure 2. The order payload as a broker receives it.

Figure 2. The order payload as a broker receives it. Each field answers what, where, how much, at what price, and under what terms. No field carries the cause. The payload is a complete instruction; it is not a record of causation.

Everything from the broker onward is tagged, empanelled, and traceable under SEBI's framework. Everything before the broker is not addressed by it. The device on which a signal is received is the only position that sits between an unanchored origin and the first point the framework reaches, the only position, in the entire sequence from signal to settlement, at which origin is still observable at all. Once an instruction leaves the device, it is a payload of symbol,

quantity, price, and side; nothing carried forward from that point can be traced back to what preceded it.

No framework built to regulate brokers can capture what brokers are never sent. This is worth stating plainly, once: the gap sits outside the regulated perimeter, not inside it. A stricter version of the same framework, tighter IP controls, deeper empanelment scrutiny, harsher penalties for unregistered algo providers, would still not reach it, because none of those measures change what arrives at the broker's API. They govern the channel more strictly. They do not extend the channel further upstream.

Evidence the gap is exploited

SEBI's own enforcement record demonstrates this gap in practice, not merely in principle. Interim Ex-Parte Order WTM/KV/ISD/ISD-SEC-7/32413/2026-27, issued 22 May 2026 by Whole Time Member Kamlesh C. Varshney, runs to 234 pages under the title "In the matter of trading in certain scrips through stock recommendations given on social media platforms."

The order names seven noticees, led by Hemant Gupta, and finds that recommendations posted through two X accounts, @WealthSolitaire, with approximately 13,600 followers as of 24 January 2026, and @desiwallstreet, with approximately 40,500 followers as of 22 January 2026, a combined reach on the order of 54,000, were amplified through a network of Telegram and WhatsApp groups to drive accumulation and distribution in illiquid SME stocks. SEBI impounded wrongful gains of ₹20,25,20,234.83.

The analytical point is narrow and does not require speculation about intent. SEBI's framework identifies, precisely, who placed the resulting orders and through which broker. It does not identify what instructed them. The order itself was built by investigators reconstructing the signal chain after the fact, cross-referencing social media posts, messaging-group membership, and trading records across an extended investigation covering the period the recommendations were posted. That reconstruction was necessary specifically because no mechanism captures the signal chain at the time the orders were placed. Enforcement, however effective after the fact, is a substitute for a capture mechanism that does not exist upstream of the broker, not an instance of one. It arrives after the loss has already been distributed among the traders who acted on the recommendation, not before.

The scale of retail exposure to this gap is documented separately, and at a different order of magnitude from any single enforcement action. In reply to Lok Sabha Unstarred Question No. 2380, answered 8 July 2025, the Ministry of Finance reported that individual traders' net losses in the equity derivatives segment rose to ₹1,05,603 crore in FY25, an increase of approximately 41 percent over FY24, with 91 percent of individual F&O traders recording a loss. This figure describes losses across the entire retail derivatives segment, not losses attributable to any one signal source or enforcement matter; it is cited here to establish that the population exposed to unaccounted-for signal origin is not a handful of noticees in a single order, but the great majority of a segment numbering in the tens of lakhs of traders.

The same absence, examined elsewhere

The absence documented above was checked against three other frameworks and one systemic report. Each is presented on its own terms; no claim is made that the pattern is universal beyond what was examined.

Article 17 of MiFID II itself requires an investment firm engaged in algorithmic trading to notify its competent authority, and to arrange for records sufficient to allow that authority to monitor compliance, including, for high-frequency trading, accurate and time-sequenced records of every order placed, cancelled, executed, or quoted. Its complete text, all six paragraphs, was read in full; the word “signal” does not appear.

The regulation implementing Article 17, Commission Delegated Regulation (EU) 2017/589, known as RTS 6, goes further on responsibility, requiring a firm to remain fully responsible for its obligations “where it outsources or procures software or hardware used in algorithmic trading,” and to always be in a position to know which trading algorithm, trader, or client is responsible for a given order. Its complete text, twenty articles, was also read in full; the word “signal” does not appear there either.

The Virtual Assets Regulatory Authority’s Technology and Information Rulebook requires a firm using algorithms to document their design, testing, and the assumptions and biases underlying them. It does not require capture of the source that informed those assumptions.

The Dubai Financial Services Authority’s Markets framework, at AMI 5.7.3, requires that direct electronic access be structured so that a firm can identify the orders and trades placed through it. It does not extend to what prompted a client to place them. These two provisions were examined at the sections directly relevant to algorithmic and electronic trading; neither rulebook was swept in full, and this section’s finding should be read with that narrower scope in mind.

The Financial Stability Board’s October 2025 report on AI adoption in the financial sector, read in full, is substantially concerned with model explainability and third-party concentration risk. The word “signal” appears once in the entire document, and refers to consumer complaint volume as an indicator of fraud, unrelated to a trading signal’s origin. It is cited here only as adjacent context on AI governance, not as support for anything argued above.

Four frameworks, one systemic report, the same absence. The reader is left to draw the wider conclusion.

What would falsify this argument

Either an order-placement interface can carry the origin of the signal that produced an order, in which case the field carrying it should be identifiable in a published broker API specification. Or it cannot, in which case no broker-side mandate, however thoroughly constructed, closes the gap described here.

This paper has looked for that field in the traceability provisions of SEBI’s framework, examined above in full. It has not found it. Any reviewer able to identify such a field in a published, production order-placement specification would falsify this paper’s central claim, and should do so on that basis alone.

3. Implications, Counter-Arguments, and Policy

A. The cost of the gap: reconstruction versus capture

The gap described in the preceding section is not merely unobserved. Its existence imposes a specific method on the regulator, and that method is visible across SEBI's own enforcement record.

Interim Ex-Parte Order WTM/KV/ISD/ISD-SEC-7/32413/2026-27, issued 22 May 2026 against seven noticees led by Hemant Gupta, arose from recommendations posted through two X accounts, @WealthSolitaire, with roughly 13,600 followers, and @desiwallstreet, with roughly 40,500, amplified through Telegram and WhatsApp groups over an extended period before SEBI impounded wrongful gains of ₹20,25,20,234.83. The order that resulted runs to 234 pages of reconstruction.

The pattern is not isolated to that matter. In the case of Yash Garg, proprietor of Yash Trading Academy, SEBI found that between 1 November 2019 and 10 April 2023 he collected ₹92,98,405.56 from investors through Telegram channels offering paid trading calls and account-handling services, marketed with claims of guaranteed returns. A show-cause notice followed on 14 November 2025; the order barring him from the securities market for two years and directing a full refund was issued in March 2026, roughly six and a half years after the collection began. In the matter of the Telegram channels "Intraday Jackpot" and "Professional Day Trading Institute," SEBI's order (April 2026, per contemporaneous legal reporting; the order's own text could not be independently machine-read for this paper, and the figures below should be read with that caveat) barred Akshay Kumar, Mithun Sah, and Arjun Sah from the market for two years, directed a refund of approximately ₹9.02 crore, and penalised a fourth individual, Beauti Kumari, for facilitating the channel's operations. And in an earlier matter, SEBI investigated the Telegram channel Safebulls, a channel with a subscriber base of approximately 60,879, following a single complaint dated 30 September 2021, running an investigation covering January 2021 to March 2022 before ordering disgorgement of ₹28,80,825 from the channel's operator and an affiliated trading firm found to have dealt in the recommended stocks ahead of publishing them.

Four matters, four different scales, one method: complaint, followed by forensic audit, followed by bank-trail reconstruction, followed by asset freeze, followed by a market ban. In each case the loss to investors had already occurred, and in most cases had already been distributed across a base of subscribers numbering in the thousands to tens of thousands, before the reconstruction that produced the order began. The time between the triggering conduct or complaint in each matter and the order's own date is itself informative: roughly two and a half years for Safebulls, and roughly six years for Yash Garg. None of that elapsed time reflects delay or inattention on the regulator's part. It reflects the time required to build, after the fact, a record that did not exist when the conduct occurred.

This is not a matter of insufficient regulatory sophistication. At SEBI's 38th Foundation Day, on 25 April 2026, the Union Finance Minister noted that SEBI's Data Analytics and Digital Forensics Laboratory is "using advance analytics, AI/ML models to detect complex market

manipulation patterns and network-based frauds.” A capability of that description applied to order flow and trading records can detect that a pattern of trading is anomalous. It cannot detect what caused a given trade, because that causation was never recorded in the data the capability operates on. The laboratory the Finance Minister described is examining the only layer that is structurally available to it. The consequence, that four separate matters each required a complaint, followed by a multi-month or multi-year reconstruction, before any order could issue, is the direct cost of a signal chain that must be inferred from screenshots, bank narratives, and messaging-group records, rather than read from a contemporaneous account of what happened. That cost is paid in the same currency in every one of the four matters: time, during which the arrangement being investigated continues to operate.

B. Counter-arguments

“This is an enforcement problem, not an architecture problem”

The strongest version of this objection notes that SEBI already prohibits unregistered investment advice under the SEBI Act and the IA and RA Regulations, and that all four matters above were, in the end, successfully prosecuted. The framework functions; what it needs is more resources for enforcement, not a new capture mechanism.

This is correct as far as it goes, and it should be conceded that far. But prohibition is not observation. A rule against giving unregistered advice does not, by itself, create a record of who gave it or when. Every one of the four matters above required reconstruction precisely because no such record existed at the time the conduct occurred; the show-cause notices, bank-transaction narratives, and Telegram screenshots that fill each order are evidence of exactly that absence. More enforcement staff, better forensic tooling, or faster case throughput would shorten the reconstruction. None of them would remove the need to reconstruct in the first place, because none of them changes what is recorded at the moment a signal is received.

“A device-resident capture layer is surveillance”

This is the most serious objection raised against anything resembling the mechanism this paper’s central claim implies could exist, and it deserves to be treated as such rather than answered in passing. Capturing where a trading signal originates necessarily means capturing something about a communication a person received on a private device, and a regulator asked to mandate, or even encourage, software with that function should be cautious, not enthusiastic, about the request.

What can honestly be said is this: origin metadata, that a signal arrived, when, and through which channel, is a narrower category than content, and a capture mechanism confined to the former is a different privacy proposition than one that reads message bodies. But narrower is not the same as unobjectionable. Questions of what is retained, for how long, under whose custody, visible to whom, revocable by what mechanism, and enforceable against misuse by what authority are not answered by the existence of a narrower category; they are policy questions a regulator would have to settle, deliberately and in public, before any such mandate could be responsibly imposed, and they are not resolved by this paper. Nor should the burden of resolving them fall to the party proposing the mechanism. The honest position is that this objection identifies a real and unresolved design space, not a settled one, and any claim otherwise would be overreach.

“Sophisticated actors will simply bypass it”

The strongest form of this objection is also, on the evidence above, correct: anyone determined to conceal a signal’s origin can route around a capture point, or place an order manually with no signal at all. No capture mechanism defeats a determined adversary, and this paper does not claim otherwise.

What the four matters above show is different from sophisticated evasion. None of them involved actors concealing their channel from view. @WealthSolitaire, @desiwallstreet, Yash Trading Academy, Intraday Jackpot, and Safebulls operated openly, under public or semi-public names, for periods ranging from months to years, at subscriber scales in the thousands and tens of thousands. What made this viable was not sophistication in evading observation, it was the absence of anything watching that layer at all. A capture mechanism does not need to defeat a determined adversary to change this. It needs only to remove the ambient invisibility that let ordinary, unconcealed activity operate at this scale for this long before a complaint arrived.

“The retail trader will not adopt this”

Retail traders already resist compliance friction, and a further mandated step is a reasonable thing to expect will be ignored, worked around, or resented, this is a fair prediction, not a weak one, and the paper does not dismiss it.

The relevant distinction is between friction a user must act on and friction a user does not notice. A requirement that a user obtain and maintain a static IP address, discussed below, is the former: a recurring cost and a configuration step the user must complete before trading at all. A mechanism that operates without requiring user action is the latter. This is a design consideration to be weighed by whoever specifies such a mechanism, not a claim that adoption has been solved or tested at any scale.

“This paper is written by someone with a commercial interest”

This should be stated plainly rather than left for a reader to raise unassisted: the author has filed a patent covering a system in this space and has built a product. The disclosure accompanying this paper says so.

The response is not to argue the interest away but to make it irrelevant to whether the argument holds. Every citation in this paper, SEBI’s circulars, its enforcement orders, the Lok Sabha record, the EU and UAE regulatory texts, is a public document any reader can retrieve and check without relying on the author’s characterisation of it. The falsifiability test proposed in the preceding section is deliberately written against published specifications and regulatory texts, and against nothing the author alone possesses. A commercial interest is a reason to verify the sources independently. It is not, by itself, a reason to doubt what independent verification would show.

C. The inclusivity tension

At the same Foundation Day address cited above, the Finance Minister stated: ***“We need enabling frameworks for responsible financial education, but we should not tolerate the monetisation of uninformed retail investor trust for personal enrichment.”*** The same address favoured a soft-touch, principles-based regulatory approach over a heavily detailed rulebook, and

urged SEBI to “invest very substantially in public awareness on every major platform in regional languages,” a call to reach retail participants beyond urban, English-speaking audiences. Separately, in reply to Lok Sabha Unstarred Question No. 2380, the Ministry of Finance reported individual F&O traders’ net losses at ₹1,05,603 crore in FY25, up roughly 41 percent year-on-year, with 91 percent of individual F&O traders recording a loss.

Set against this is a specific requirement within the algorithmic trading framework examined in the preceding section: to trade algorithmically through a broker’s API, a retail participant’s access must run over a static, whitelisted IP address. A static IP is not typically part of a standard retail internet connection in India; obtaining one is ordinarily a recurring cost and a technical configuration step, and it is a step disproportionately more familiar to an urban, technically literate user than to the broader, regional-language audience the same address called on SEBI to reach.

This paper states the tension and does not resolve it. A compliance requirement that a user must purchase and configure is not, by construction, friction-free, regardless of the soundness of its underlying purpose. A mechanism that requires no comparable action from the user is a different kind of requirement, in this one specific respect. What follows from this observation is a question of policy design, how a jurisdiction balances traceability requirements against the friction they impose on the population it also wants to broaden, and this paper does not answer it. It notes only that the two stated objectives, broader and more inclusive participation on one hand and a static-IP-anchored traceability requirement on the other, pull in different directions for exactly the segment the Finance Minister’s address named as the priority to reach.

Whether these frictions are inherent to the requirement, or artefacts of where in the chain it is imposed, is an empirical question. Section 4 addresses it.

D. What closing the gap would enable

Three implications follow, stated as consequence rather than recommendation.

First, audit would become reading rather than reconstruction. A signal chain available at the time a trade is placed replaces a forensic exercise that, in the matters examined above, took months to years per case.

Second, registration would become verifiable rather than inferable. Where origin is observable, the distinction between a recommendation from a SEBI-registered Research Analyst or Investment Adviser and one from an unregistered channel becomes something that can be checked at the point of trade, which is a precondition the existing IA and RA Regulations already assume but cannot currently confirm before the fact.

Third, enforcement would shift from a retrospective exercise to a structural one. Not because evasion would become impossible, the counter-argument above concedes it would not, but because the default condition of a signal reaching a trade would change from unobserved to observed.

4. Empirical Evidence

This section proves architecture, not scale. One complete trace establishes that the mechanism operates. Two traces, one rejected, one placed, establish both branches of it. Nothing that follows adds to that proof by adding volume; it only shows the same proof twice, in the two outcomes an order can have.

The claim is exactly this: a fully compliant order, valid Algo ID, whitelisted static IP, empanelled provider, reaches the broker while its signal origin remains structurally invisible to that broker. The origin was captured, at the device, before the order existed. It is present in the record. It is absent from everything downstream. What follows demonstrates this in production, under the mandate examined earlier in this paper.

Finding 1: the provenance gap, demonstrated

A signal enters the system and is stamped with its origin before anything else happens to it. A production log line, reproduced here in full and unredacted at the one field that matters most:

```
2026-06-01 14:16:02,904 | INFO | websocket_listener |
WSS_SIGNAL_RECEIVED
| source=wss://dashboard.aiondashboard.site/ws/alerts
| symbol=SILVERM26JUN269000PE | pattern=bennys_signal
| received_at=2026-06-01T08:46:02.904800+00:00
```

The source field is not redacted here, and should not be. It is the exhibit. What it shows is a signal, timestamped at receipt, carrying a record of where it came from, before any risk evaluation, before any broker is involved, before an order exists in any form.

From that point, one of two things happens. A rejected case, drawn directly from the production order-attempt log:

```
2026-04-24T08:21:52Z | symbol=HDFCBANK | pattern=volume_breakout
| signal_origin_id=NSEHDFCBANK_1777018853387_178b3399
| outcome=REJECTED | reason="Mandatory SL/TP missing for auto-executed
order"
```

And a placed case, from the same log:

```
2026-04-24T09:59:41Z | symbol=GOLDM26APR150500PE |
pattern=order_flow_breakout
| signal_origin_id=MCXGOLDM26APR150500PE_1777024777160_55698984
| outcome=PLACED | broker_order_id=...7152
```

The signal-origin identifier is present in both rows, regardless of outcome. In the placed case, the order that reached the broker satisfies everything the framework examined earlier in this paper requires: it carries a valid Algo ID, it travels over a whitelisted static IP, it is placed by an empanelled provider. The broker’s record of it will show who placed it, through which channel, under which credentials. Nothing in that record shows what caused it. The cause, the signal, its timestamp, its stated source, exists only in the device log reproduced above, one step removed from the broker’s view and never transmitted to it.

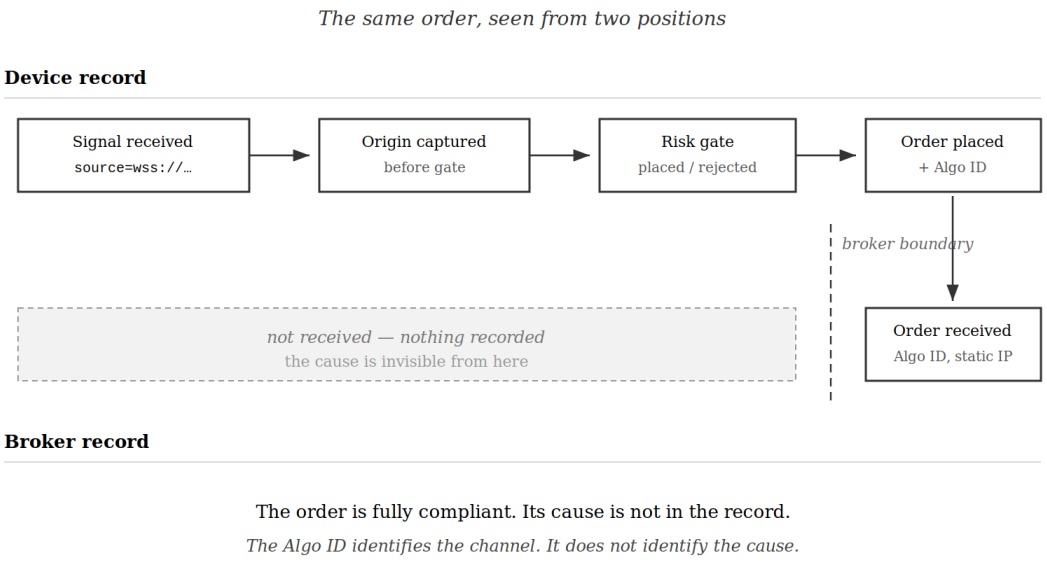


Figure 3. The same order, seen from two positions.

Figure 3. The same order, from two positions. The device records the signal, its source, and the gate decision. The broker receives a compliant order, valid Algo ID, whitelisted static IP, empanelled provider, and no record of what caused it.

The Algo ID identifies the channel. It does not identify the cause. That is the entire finding, and it is demonstrated, not argued, by the two rows above.

Note what the signal-origin identifier does in the rejected row as much as the placed one: it persists regardless of what the risk gate decides. The gate evaluates the order; it does not touch the record of where the order came from. That separation, origin capture happening before and independent of the compliance decision, is what makes the record above a record of signals rather than a record of trades. A system that only logged placed orders would show nothing about the rejected one at all, including the fact that it had a traceable origin in the first place.

The two traces above are not a one-off demonstration. The listener that produces rows in this shape has logged 54,148 such records between 17 June and 13 July 2026, and was still appending at the time of writing. The mechanism the traces illustrate is not a single captured instance; it is a continuously operating part of production infrastructure over that window. A fuller extract, correlating captured signals with their eventual order outcome, is presented in Appendix A.

Finding 2: local risk enforcement

A user sets a daily loss ceiling, a profit target, or both, through the system’s interface. The agent enforces this locally as a hard stop that does not reset until the next trading session begins. This requires nothing from the broker and nothing from the regulator; it is enforced entirely at the point where the signal is already being evaluated.

The rejected trace above is itself an instance of the gate holding: an order without mandatory stop-loss and target protection is refused before it reaches the broker, rather than placed and left

unprotected. The same gate structure, refuse rather than expose an unbounded position, is what a daily loss or profit-target breach also triggers.

What is claimed here is narrow and should not be read as more than it is: the mechanism exists, and the record shows it holding. No claim is made that this mechanism reduces trading losses in aggregate; no outcome data in this record supports that comparison, and none is offered. The retail loss figures cited earlier in this paper describe a market-wide population under no such mechanism. This paper does not draw a line between the two. A reader may.

Finding 3: privacy by design

Users in this record are represented by a non-reversible identifier. Identity is not retrievable from the operational data path. A user count is therefore not reported as a headline metric in this section, not because it would be inconvenient, but because the architecture is deliberately blind to it. What the system evidences is not who traded, but where the signal came from, which is precisely the distinction this paper argues the existing regulatory framework does not reach. The absence of an identity trail here is not a gap in this evidence. It is the same distinction the rest of this paper makes, demonstrated in the shape of the record itself.

Finding 4: inclusion by design

India’s national position on artificial intelligence and digital infrastructure is explicitly inclusive. The stated objective, repeated at the highest level, is that the benefits of this infrastructure must reach beyond urban, English-speaking, technically literate populations.

At SEBI’s 38th Foundation Day, the Union Finance Minister urged the regulator to invest substantially in public awareness on every major platform, in regional languages, so that investor protection is not confined to that narrower audience. The same address favoured a soft-touch, principles-based approach and urged simplification to reduce friction.

The algorithmic trading framework examined in Section 2 became fully mandatory on 1 April 2026. To trade algorithmically, a retail participant’s access must run over a static, whitelisted IP address.

A static IP is a property of fixed-line broadband. India’s internet population, 1.03 billion users at the end of 2025, a penetration rate of 70.0 percent, alongside 1.06 billion active cellular connections, equal to 72.5 percent of the total population, connects predominantly by mobile. The Telecom Regulatory Authority of India separately reported that, as of March 2026, wireless connections accounted for 1,046.26 million of India’s 1,092.79 million internet subscribers, with wired connections accounting for the remaining 46.54 million, wireless subscriptions representing approximately 96 percent of the total. A mobile connection does not carry a static IP by default. Obtaining one is a paid addition to a retail connection, where it is available at all; a published, verifiable tariff figure for this addition was not identified during this research, and this paper states only the direction of the finding, not a price.

The requirement therefore falls most heavily on precisely the population the stated objective seeks to include. This is not a criticism of the framework’s design, which addresses a real traceability need. It is an observation that a traceability requirement imposed at one layer of the

chain produces an access cost at another, and that the cost is not evenly distributed across the population the framework governs.

Whether that cost is inherent to the requirement, or an artefact of where in the chain it is imposed, is an empirical question. What follows addresses it.

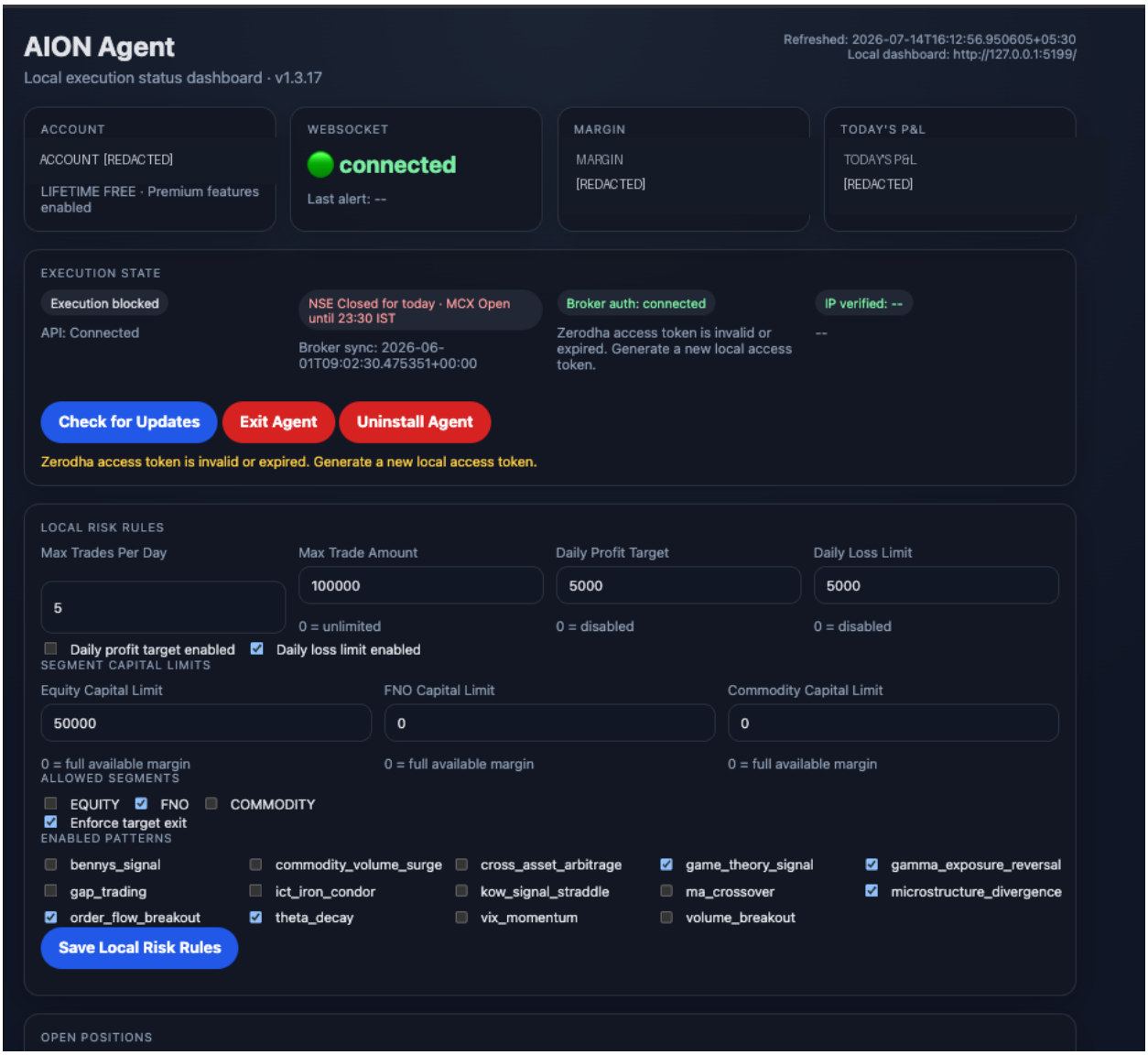


Figure 4. The agent interface.

Figure 4. The agent interface. The static-IP requirement is detected and reconciled by the system, not configured by the user. Broker authorisation is a single action. Risk limits are expressed in rupees, a daily trade ceiling, a daily profit target, and a daily loss limit, not in quantitative parameters.

Three observations follow from the interface shown.

First, the interface resolves the static-IP requirement at the device rather than asking the user to satisfy it directly. It detects the device's current public address, compares it against the address registered with the broker, and reports whether the two are aligned. Beyond an initial, single broker-authorisation step, the user is not asked to configure a network setting or to understand what a static IP is in order to satisfy the requirement.

Second, risk is expressed in terms the user already possesses. The interface asks how many trades are permitted per day, what the largest single trade may be, at what daily profit the system should stop, **and at what daily loss it should stop,** each expressed in rupees or as a count, not as a volatility parameter or a Greek. A participant who cannot price an option or read a volatility surface can still answer all three questions, because each concerns the participant's own tolerance rather than the market. The mechanism does not require the participant to possess quantitative knowledge. It requires them to know what they can afford to lose.

Third, the participant's own exposure is visible to the participant, and to no one else. Positions and the day's profit and loss are synchronised and displayed on the device in real time; this data is held locally and is not transmitted, and balances do not appear in the telemetry examined in Finding 1. The participant gains visibility over their own exposure without surrendering that visibility to anyone else, the same privacy property established in Finding 3, expressed here as something the user directly experiences rather than as a property of the system's design.

This mechanism assumes the participant's signal provider is registered with the regulator. This paper does not claim it makes an unregistered source safe to follow. It claims that origin becomes observable, which is the precondition for the registration requirement examined in Section 2 to be enforceable at the point of trade at all, rather than only after an investigation has reconstructed it.

None of this is a claim that the interface increases participation, that it has been tested among tier-2 or tier-3 city users, or that it reduces trading losses; no adoption or outcome data of that kind exists for this record. The claim is narrower: the specific frictions identified in Section 3C, a device class the requirement assumes, and knowledge the requirement implicitly demands, are addressable at the device layer. That is the entire claim this finding makes.

Scope and method

This record was produced on the author's own infrastructure, using the author's own broker accounts, on a local machine. No external user traded on this system, and the signal source shown above is the author's own relay, not a third party.

This is stated as a design choice, not a limitation to excuse. The capture mechanism stamps the origin of whatever transport delivers a signal; it does not depend on that transport being any particular provider. Demonstrating it against a live, unregistered third-party signal channel operating on real external traders would have required running exactly the kind of unregistered platform this paper's regulatory analysis is concerned with. The author did not take that route. Live order placement ceased on 1 June 2026. Signal generation and origin capture continue for research purposes: no capital is committed, no orders are placed, and no external party trades on the system. Controlled evidence, produced this way, is what evidence gathered without causing the harm under discussion looks like.

The alternative, deferring publication until the mechanism could be shown against an external, unregistered provider at scale, is not a more rigorous standard. It is a request to first operate the unregistered platform this paper's own regulatory analysis identifies as the source of the harm, in order to document it. The evidence presented here does not require that trade-off, and was not produced by making it.

Limitations

Four apply, and only these:

First, capture is demonstrated; verified attribution is not. The origin record carries the fields `registered: false` and `signature_verified: false` in this data. A provider-registry check and a cryptographic signature verification of the captured identity were not implemented. This paper's claim is that origin is captured and preserved. It is not a claim that the captured identity has been verified against any registry, and no such claim should be inferred from it.

Second, this record is author-held. Unlike the regulatory citations elsewhere in this paper, it cannot be independently retrieved by a reader. The redacted record is available for inspection by a regulator or an independent reviewer on request.

Third, the record covers a single broker under a single jurisdiction's framework, as examined throughout this paper. No wider claim is made.

Fourth, this is a feasibility demonstration. It establishes that the mechanism operates. It does not establish behaviour at market scale, and none is claimed.

Note on redaction

The traces above include the signal source URL, symbol, pattern, timestamps, outcome, rejection reason, and the last four digits of the broker order ID, none of which are redacted. Account identifiers, IP addresses, balances, margins, profit-and-loss figures, device fingerprints, and user-agent strings have been removed entirely and do not appear in either trace, or in Figure 4. The underlying production record also contains client IP addresses associated with system access; none appear in this or any other extract in this paper.

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5. A General Form

The gap examined in this paper has a general shape, and stating it plainly may be more useful than the specific case.

Every regulatory framework must attach accountability at an observable point. In securities markets, that point is the registered intermediary, the broker. It is chosen because it is the one participant every transaction must pass through, and the one whose conduct a regulator can compel.

But a chokepoint is not a vantage point. A framework anchored at an intermediary can observe everything the intermediary receives. It cannot observe what the intermediary was never sent. Where the conduct of interest occurs upstream of the anchor, the framework is structurally blind

to it, not through poor drafting, but because the instrument it regulates does not carry the information required.

The failure mode that follows is therefore predictable, and it is not unique to securities:

> When a framework anchors accountability at an intermediary, and the conduct of regulatory interest occurs upstream of that intermediary, the framework will reconstruct rather than observe. Enforcement becomes forensic. It arrives after the harm, at a cost measured in years.

This suggests a diagnostic question, stated here as the paper's method because it is generalisable beyond the case examined:

> For any framework: what does the regulated entity actually receive? And is the conduct of concern present in what it receives?

If the answer to the second question is no, no strengthening of obligations on that entity will close the gap. The obligation must attach at a point where the conduct is observable, or the conduct must be made observable at the point where the obligation attaches. The second option is what this paper's evidence, in Section 4, demonstrates is possible.

The specific gap examined here concerns trading signals. The general form concerns any regulated chain in which causation precedes the first accountable party. As algorithmic and AI-mediated decision-making moves further upstream of the entities that frameworks are able to reach, the distance between the point of causation and the point of accountability is likely to widen rather than narrow. The diagnostic above is offered on the expectation that this case will not be the last of its kind.

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Appendix A, Redacted Production Record

The following extract is drawn from the production record described in Section 4. It corrects an error from an earlier draft, which labelled a per-signal identifier as the signal source. The two are different things, and this appendix separates them:

- The **signal source** is the origin: the WSS relay address from which the signal was received. It is what the broker never receives, and it is the exhibit.
- The **linkage timestamp** is the join key used to correlate a captured signal with its eventual order outcome. This production system does not carry a single shared identifier across the signal-receipt log and the order-execution log; the two are correlated here by matching symbol and sub-second timestamp proximity within the same continuous log stream, which this paper treats as a verifiable, though narrower, form of linkage than a shared database key. Rows for which this correlation could not be established have been excluded rather than approximated.

Because of this stricter standard, this extract is smaller than a prior draft and covers a narrower window (1 June 2026) and a single exchange segment (NFO). Account identifiers, IP addresses, and all financial values have been removed and appear nowhere in this extract. The full production record is available for inspection by a regulator or independent reviewer on request.

Order timestamp (IST)	Exchange	Symbol	Pattern	Signal source	Linkage timestamp (signal received_at, UTC)	Outcome
2026-06-01T12:45:25	NFO	NIFTY26600224000CE	gamma_exposure_reversal	wss://dashboard.aiondashboard.site/ws/alerts	2026-06-01T07:15:23.939121+00:00	GTT_FAILED, forced exit

Order timestamp (IST)	Exchange	Symbol	Pattern	Signal source	Linkage timestamp (signal received_at, UTC)	Outcome
2026-06-01T13:15:46	NFO	NIFTY2660224150CE	gamma_exposure_reversal	wss://dashboard.aiondashboard.site/ws/alerts	2026-06-01T07:45:45.865599+00:00	GTT_FAILED, forced exit
2026-06-01T14:06:46	NFO	NIFTY2660223750CE	gamma_exposure_reversal	wss://dashboard.aiondashboard.site/ws/alerts	2026-06-01T08:36:45.914112+00:00	GTT_FAILED, forced exit
2026-06-01T14:15:04	NFO	NIFTY2660223700CE	gamma_exposure_reversal	wss://dashboard.aiondashboard.site/ws/alerts	2026-06-01T08:45:04.028593+00:00	GTT_FAILED, forced exit
2026-06-01T14:25:50	NFO	NIFTY2660224150CE	game_theory_signal	wss://dashboard.aiondashboard.site/ws/alerts	2026-06-01T08:55:47.703882+00:00	ENTRY_SUCCESS, GTT_SUCCESS, order ID … 0256

Four of the five rows resulted in an entry followed by a failed protective-bracket placement and a forced exit, a distinct, and separately logged, outcome from either a clean placement or a pre-trade rejection; this paper's earlier findings describe the pre-trade rejection and clean-placement branches, and this outcome is included here because it appears in the genuinely correlated data for this window, not selected to represent any particular rate of occurrence. The fifth row is a clean, fully successful placement for the same underlying symbol at a later signal instance. No claim is made about the proportion of outcomes of each type across the wider record; this extract demonstrates that the source-to-outcome link holds across more than one outcome type, not the frequency of any of them.